

1. Motivation

Geopolitical crises between major powers — such as the U.S.–Iran tensions of the late 2010s and early 2020s — have repeatedly demonstrated how quickly political conflicts translate into economic disruption. When diplomatic relations deteriorate, oil supply chains are threatened, Brent Crude prices spike, and the knock-on effects ripple through inflation, interest rates, and household purchasing power across the globe. These are not abstract macro phenomena: they change what people pay for groceries, heating, and transport within weeks of a political event.

The central question motivating this project is: do countries that have invested in renewable energy infrastructure absorb these oil price shocks better than those that remain dependent on fossil fuel imports? In other words, does the green energy transition serve as an economic buffer against geopolitical volatility? This is a policy-relevant question with direct implications for how governments should prioritize energy investment, and it sits at the intersection of energy economics, international relations, and data science.

To answer this question, we built a full data science pipeline spanning exploratory analysis, statistical hypothesis testing, and machine learning — applied to a panel of 91 countries observed annually from 2000 to 2020.

2. Data Source

This project uses three publicly available datasets, merged into a single panel dataset.

Primary Dataset The Global Data on Sustainable Energy (2000–2020), published jointly by the IEA and the World Bank, provides annual observations for 173 countries. From this dataset we use: renewable energy share (% of total energy consumption), GDP growth (annual %), and GDP per capita (current USD).

Enrichment Datasets

- Brent Crude Oil Prices: daily price series, aggregated to annual averages. Year-on-year percentage change is computed as the `oil_price_change` feature.
- World Bank Open Data — Energy Import Dependency: annual share of energy consumption sourced from imports (`energy_import_pct`), covering the same country-year panel.

An interaction term `renewable_x_oil` was engineered as the product of `renewable_share` and `oil_price_change`, capturing whether high renewable adoption modulates the effect of oil price shocks on growth — the core hypothesis of the project.

Final Dataset Properties

- Observations: 1,473

- Countries: 91
- Time period: 2000–2020
- Features: renewable_share, gdp_per_capita, energy_import_pct, oil_price_change, renewable_x_oil
- Targets: gdp_growth (continuous), gdp_growth_class (binary: 1 if growth > 0)
- Class balance: ~90% expansion, ~10% contraction

All datasets are freely available in CSV format. Data was merged on (country, year) keys with inner join, reducing the 173-country base to 91 countries with complete observations across all three sources.

3. Data Analysis

3.1 Exploratory Data Analysis Descriptive statistics revealed that GDP growth averages 3.57% across the panel (std = 3.69), with a minimum of -17.67% (crisis years) and renewable share ranging from 0.26% to near-100%. GDP per capita spans from very low-income to high-income economies (mean ~\$15,600, std ~\$20,300), indicating high cross-country heterogeneity.

A correlation matrix showed that all five features have statistically significant associations with GDP growth. Visual inspection via scatter plots confirmed that the relationship between oil price change and GDP growth is positive but noisy, and that high-renewable countries tend to cluster at higher GDP per capita levels — suggesting a confounding structure that purely bivariate analysis cannot resolve.

Grouping countries by renewable share tercile (Low: 0–25%, Medium: 25–50%, High: >50%) showed a visible difference in average GDP growth stability across groups — high-renewable countries exhibited lower variance in growth rates, particularly around crisis years.

3.2 Hypothesis Testing Four hypotheses were tested using Pearson and Spearman correlation tests (n=1,473). All four were statistically supported at $p < 0.001$:

- H1: Oil price change → GDP growth | Pearson $r=0.237$, Spearman $r=0.226$ | Supported
- H2: Renewable share → GDP growth | Pearson $r=0.171$, Spearman $r=0.216$ | Supported
- H3: Interaction (renewable $\times$ oil) → GDP growth | Pearson $r=0.177$, Spearman $r=0.264$ | Supported
- H4: Energy import pct → GDP growth | Statistically significant | Supported

A fixed-effects OLS model (controlling for country and year fixed effects) achieved $R^2=0.447$, with `oil_price_change` as the strongest coefficient (0.041, $p<0.001$). The interaction term `oil_price_change × renewable_share` had a negative coefficient (-0.0005), consistent with H3: higher renewable share slightly dampens the pass-through of oil shocks to GDP growth.

3.3 Multicollinearity Check (VIF) All five features have VIF scores below 2.5, ruling out multicollinearity. No feature needed to be dropped.

3.4 Data Preprocessing 80/20 train/test split (1,178 train, 295 test). Classification split was stratified. All features standardized with `StandardScaler` (mean=0, std=1).

3.5 Machine Learning Pipeline

Regression (predicting continuous GDP growth): Random Forest was tuned with `RandomizedSearchCV` (20 iterations, 5-fold CV). Best config: `n_estimators=100`, `max_depth=10`, `min_samples_leaf=2`. XGBoost trained with fixed hyperparameters (`n_estimators=300`, `learning_rate=0.05`, `max_depth=5`).

Results:

- Random Forest: RMSE=3.396, MAE=2.155, $R^2=0.267$
- XGBoost: RMSE=3.585, MAE=2.271, $R^2=0.183$

Unsupervised (grouping countries): K-Means with $K=3$ (selected via elbow method and Silhouette score=0.298). PCA reduced features to 2 components capturing 61.1% of variance. Three groups emerged: high-income energy-secure economies, transition economies, and high-import dependent economies.

Classification (expansion vs. contraction): Logistic Regression with L2 regularization, `class_weight='balanced'`.

- Precision (contraction): 0.09 | Precision (expansion): 0.90
- Recall (contraction): 0.45 | Recall (expansion): 0.53
- Macro F1: 0.41 | Accuracy: 0.52 | ROC AUC: 0.518

4. Findings

4.1 Energy features predict GDP growth direction, but weakly Random Forest achieves $R^2=0.267$ on the test set — our five features explain ~27% of variance in GDP growth. The fixed-effects model with controls reaches $R^2=0.447$, confirming real signal but also large unexplained variance.

4.2 GDP per capita and oil prices are the dominant drivers Feature importance (Gini): `gdp_per_capita` (0.33) > `oil_price_change` (0.21) > `renewable_share` (0.19) >

energy_import_pct (0.16) > renewable_x_oil (0.11). Development level is the strongest predictor; oil prices rank second.

4.3 Renewable share matters, but as part of a system The interaction term supports H3: renewable adoption partially dampens the GDP impact of oil price volatility. However, this effect is moderate and entangled with income level — high-renewable countries are also typically high-income.

4.4 Recession prediction is genuinely hard with energy data alone Logistic Regression: macro F1=0.41, ROC AUC=0.518, contraction precision=0.09. Energy features alone cannot predict recessions reliably — contractions are driven by financial crises and geopolitical shocks not captured by annual energy statistics. This is an honest negative result.

4.5 Three distinct country typologies K-Means identified: Green Growth Leaders (high renewable, high GDP per capita, highest expansion probabilities), Vulnerable Importers (high energy_import_pct, lowest expansion probabilities), and Energy-Independent Economies (net exporters/self-sufficient, variable trajectories).

5. Limitations and Future Work

Limitations

- Missing confounders: monetary policy, fiscal stimulus, demographics, and institutional quality are absent — likely explaining much of the remaining 73% unexplained variance.
- Annual resolution: oil price shocks affect economies within weeks; annual data misses short-run dynamics.
- Binary target collapses information: a country growing at 0.1% and one at 8% are both labeled "expansion."
- Income confounding: high-renewable and high-GDP-per-capita countries overlap heavily, making causal attribution difficult.
- Dataset ends in 2020: the 2021–2022 energy crisis and Russia–Ukraine war are not included.

Future Work

- Apply causal inference methods (instrumental variables, difference-in-differences) to isolate the renewable buffer effect.
- Use higher-frequency data (quarterly GDP, monthly oil prices) to capture short-run dynamics.
- Extend to post-2020 to include the 2021–2022 energy crisis as a natural experiment.

- Tune XGBoost with RandomizedSearchCV for a fair comparison with Random Forest.
- Try LSTM or time-series models that account for temporal autocorrelation in GDP growth.

AI Usage Disclosure: Claude (Anthropic) was used for code review, report writing, and notebook structuring. All analytical decisions, dataset selection, hypothesis formulation, and model choices were made by the student.